Mini Project

Solar reference access 2035 moderate supply curve

1. Project Overview and Objective

This Project focuses on analysing the Solar Reference Access 2035 – Moderate Supply curve dataset using Excel & Power BI. The dataset contains information about solar resource potential, land availability, installed capacity, generation cost (LCOE – Levelized Cost of energy), transmission cost, and total energy cost across various locations.

Objective:

The main objective of this project is:

- To analyse solar energy potential across regions.
- To evaluate cost feasibility of solar power generation.
- To identify cost-efficient and high-capacity solar sites.
- To build an interactive Power BI dashboard for energy investment decision-making.

---

2. Data Sources

- **Source Description and Timeline:** https://data.openei.org/files/8319/solar_reference_access_2035_moderate_supply_curve.csv
- **Supply Curve Type:** Moderate
- **Time Horizon:** Projection for year 2035
- **Domain**: Renewable Energy

---

3. Problem Statement

- To analyse solar resource availability and capacity potential across different locations.
- To examine how transmission distance and grid reinforcement affect total energy cost.
- To identify sites with lowest LCOE (Levelized Cost of Energy).
- To evaluate investment-ready solar locations based on cost and efficiency.

---

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Site ID	Integer	Unique identifier for each solar site location
Latitude & Longitude	Decimal	Geographic coordinate (North-South position) Geographic coordinate (East-West position)
Country	Catagorical	Country where the site is located
State	Catagorical	State/ Region name
County	Catagorical	District/ Administrative Region
Time Zone & Elevation (Mtr)	Numeric	Height above sea level
Solar Resource (kWh/m²/day)	Numeric	Average solar irradiance received
Regional Capacity Multiplier	Decimal	Adjustment factor for regional efficiency
Developable Area (Sq.Km)	Numeric	Land available for solar development
Installed Capacity (AC-MW)	Numeric	Usable grid-connected capacity
Installed Capacity (DC-MW)	Numeric	Raw solar panel capacity
Capacity Factor (AC %)	Decimal	Actual AC energy production ratio
Capacity Factor (DC %)	Decimal	DC production efficiency
Generation Cost LCOE (USD/MWh)	Numeric	Cost to generate electricity
Distance to Grid (Km)	Numeric	Distance to nearest grid connection
Grid Reinforcement Distance (Km)	Numeric	Distance requiring grid strengthening
Grid Upgrade Cost (USD/MW)	Numeric	Cost of grid infrastructure upgrade
Total Transmission Cost (USD/MW)	Numeric	Combined transmission cost
Transmission Cost LCOT (USD/MWh)	Numeric	Transmission cost per MWh
Total Energy Cost (LCOE + LCOT/ USD/MWh)	Numeric	Final electricity cost (Generation + Transmission)

5. Tools & Technologies

- **Excel:** Data cleaning, Transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, Visualization, and interactive dashboard creation.

---

6. Data Pre-Processing

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, remaned columns clearly, converted capacity factor in percentage, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

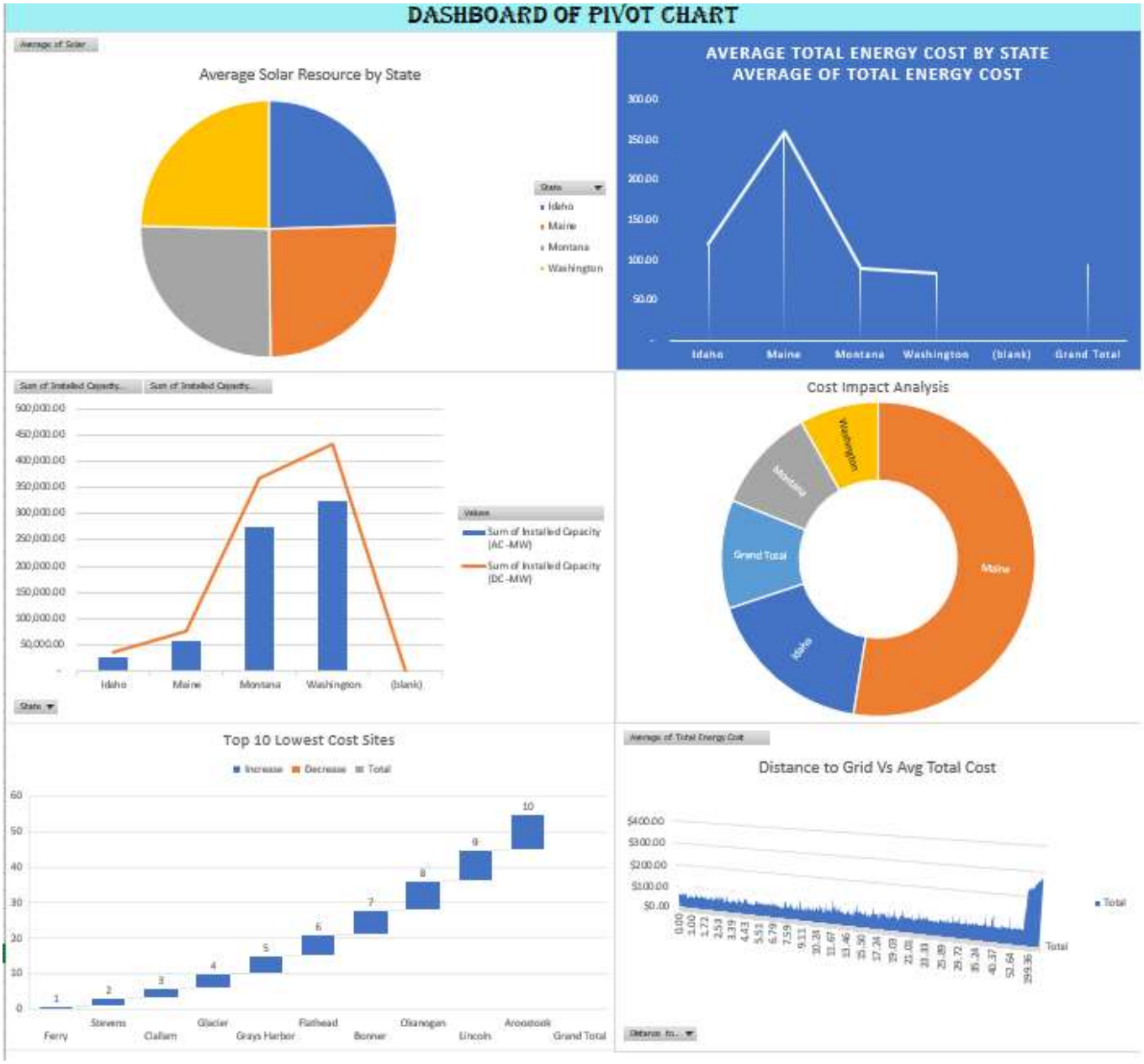
Average Solar resource by state	
State	**Average of Solar**
Idaho	3.46
Maine	3.56
Montana	3.61
Washington	3.48
Grand Total	**3.51**

Average Total Energy Cost by State	
State	**Average of Total Energy Cost**
Idaho	120.44
Maine	260.45
Montana	89.78
Washington	83.86
(blank)	
Grand Total	**94.58**

Total Installed Capacity by State		
County	**Sum of Installed Capacity (AC -MW)**	**Sum of Installed Capacity (DC -MW)**
Idaho	26,873.73	36,010.80
Maine	57,047.56	76,443.74
Montana	274,214.40	367,447.32
Washington	323,587.29	433,607.00
(blank)		
Grand Total	**681,722.99**	**913,508.87**

Cost Impact Analysis	
State	**Average of Transmission Cost**
Idaho	$70.55
Maine	$213.04
Montana	$43.75
Washington	$33.06
Grand Total	**$45.25**

Dashboard – Pivot Chart:-



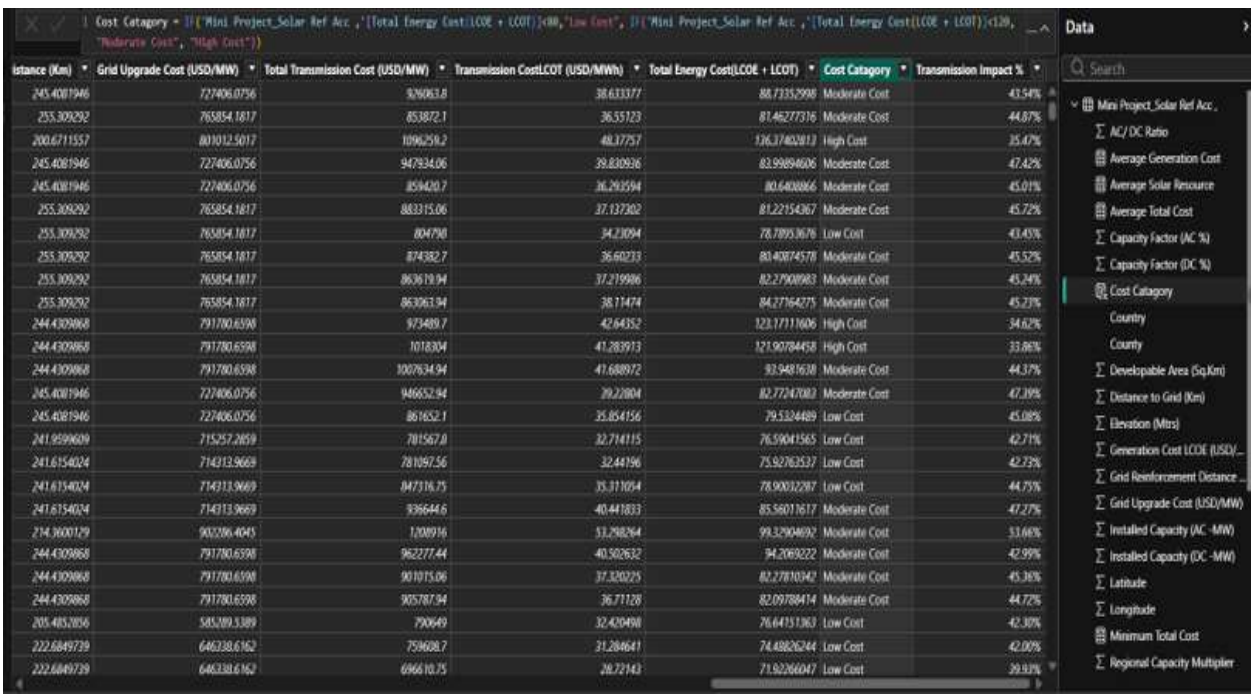
7. Data Modelling and DAX (Power BI)

- The cleaned dataset was imported into Power BI and used as a single main table for analysis. The data types of all columns were properly defined (numeric, decimal, text, and percentage). DAX measures were created to calculate key metrics such as total installed capacity, average solar resource, and total energy cost.
- The model was checked to ensure accurate calculations and proper filtering across visuals.

SCREENSHOT OF DATA MODELLING:-

DAX Columns – Cost Category

Formula: Cost Catagory = IF('Mini Project_Solar Ref Acc ,'[Total Energy Cost(LCOE + LCOT)]<80,"Low Cost", IF('Mini Project_Solar Ref Acc ,'[Total Energy Cost(LCOE + LCOT)]<120,"Moderate Cost", "High Cost"))



istance (Km)	Grid Upgrade Cost (USD/MW)	Total Transmission Cost (USD/MW)	Transmission CostLCOT (USD/MWh)	Total Energy Cost(LCOE + LCOT)	Cost Catagory	Transmission Impact %
245.4081946	727406.0756	926063.8	38.633377	88.73352998	Moderate Cost	43.54%
255.309292	765854.1817	853872.1	36.55123	81.46277316	Moderate Cost	44.87%
200.6711557	803012.5017	1096259.2	48.37757	126.37402813	High Cost	35.47%
245.4081946	727406.0756	947934.06	39.830936	83.99894606	Moderate Cost	47.42%
245.4081946	727406.0756	859420.7	36.293594	80.6408866	Moderate Cost	45.01%
255.309292	765854.1817	883315.06	37.137302	81.22154367	Moderate Cost	45.72%
255.309292	765854.1817	804790	34.23094	78.78953676	Low Cost	43.45%
255.309292	765854.1817	874382.7	36.60233	80.40874578	Moderate Cost	45.52%
255.309292	765854.1817	863619.94	37.219986	82.27908983	Moderate Cost	45.24%
255.309292	765854.1817	863063.94	38.11474	84.27164275	Moderate Cost	45.23%
244.4309868	791780.6598	973489.7	42.64352	123.17111606	High Cost	34.62%
244.4309868	791780.6598	1018304	41.283973	121.90784458	High Cost	33.86%
244.4309868	791780.6598	1007634.94	47.688972	93.9481638	Moderate Cost	44.37%
245.4081946	727406.0756	946652.94	39.22804	82.77247082	Moderate Cost	47.39%
245.4081946	727406.0756	861652.1	35.854156	79.5324489	Low Cost	45.08%
241.9599609	715257.2859	781567.0	32.714115	76.59041565	Low Cost	42.71%
241.6154024	714313.9669	781097.56	32.44196	75.92763537	Low Cost	42.73%
241.6154024	714313.9669	847316.75	35.311054	78.90032287	Low Cost	44.75%
241.6154024	714313.9669	936644.6	40.441833	85.56017617	Moderate Cost	47.27%
214.3600129	902286.4045	1208916	53.298264	99.32904692	Moderate Cost	53.66%
244.4309868	791780.6598	962277.44	40.502632	94.2069222	Moderate Cost	42.99%
244.4309868	791780.6598	901075.06	37.320225	82.27810342	Moderate Cost	45.36%
244.4309868	791780.6598	905787.94	36.77128	82.09788414	Moderate Cost	44.72%
205.4852856	585289.5389	790649	32.420498	76.64751363	Low Cost	42.30%
222.6849739	646338.6162	759608.7	31.284641	74.48826244	Low Cost	42.00%
222.6849739	646338.6162	696610.75	28.72143	71.92266047	Low Cost	39.93%

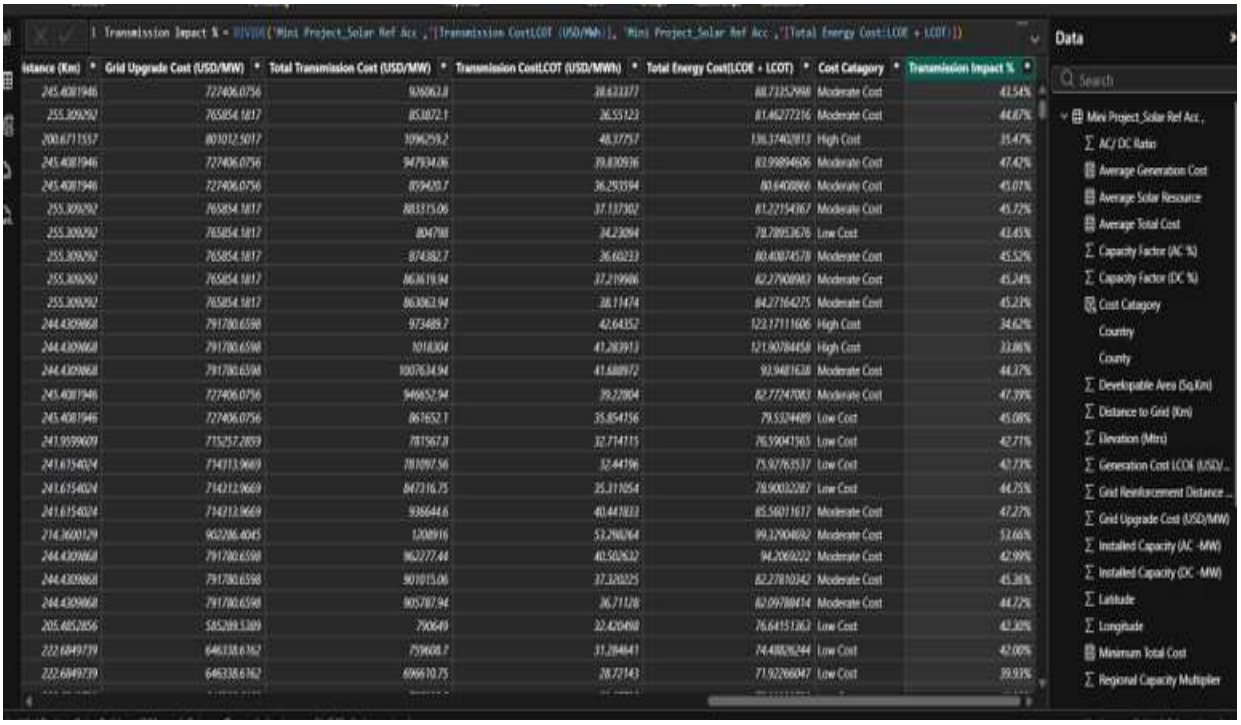
The Total Energy Cost (LCOE All-in USD/MWh) was classified into three categories:

- Low Cost – Sites with lower total energy cost (high investment potential)
- Moderate Cost – Sites with balanced cost feasibility
- High Cost – Sites with higher transmission or infrastructure burden

This categorization helps simplify cost comparison and improve decision-making.

DAX Columns: Transmission Impact %

Formula: Transmission Impact % = DIVIDE('Mini Project_Solar Ref Acc ,'[Transmission CostLCOT (USD/MWh)], 'Mini Project_Solar Ref Acc ,'[Total Energy Cost(LCOE + LCOT)])

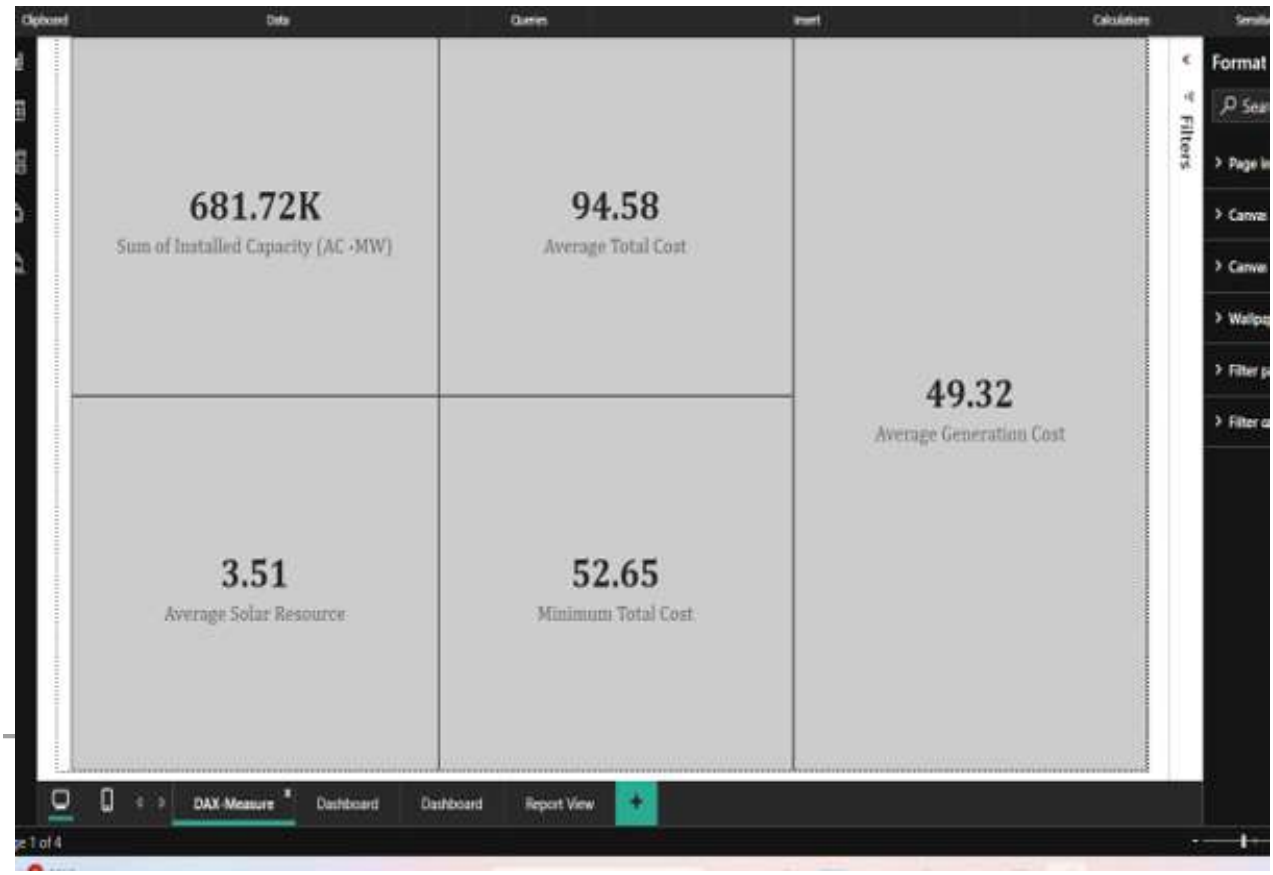


Grid Upgrade Cost (USD/MW)	Total Transmission Cost (USD/MW)	Transmission CostLCOT (USD/MWh)	Total Energy Cost(LCOE + LCOT)	Cost Catagory	Transmission Impact %
727406.0756	926062.8	28.633177	88.73352998	Moderate Cost	43.54%
765854.1817	853072.1	26.55123	81.46277216	Moderate Cost	44.67%
801012.5017	1096259.2	48.37757	136.3740813	High Cost	35.47%
727406.0756	947934.06	39.830936	83.99894606	Moderate Cost	47.42%
727406.0756	859420.7	36.293594	80.6408866	Moderate Cost	45.01%
765854.1817	883315.06	37.137302	81.22154367	Moderate Cost	45.72%
765854.1817	804788	34.23094	78.78953676	Low Cost	43.45%
765854.1817	874382.7	36.60233	80.40874578	Moderate Cost	45.52%
765854.1817	863619.94	37.219986	82.27908983	Moderate Cost	45.24%
765854.1817	863062.94	28.11474	84.27164275	Moderate Cost	45.22%
791780.6598	973489.7	42.64352	122.17111606	High Cost	34.62%
791780.6598	1018304	41.283913	121.90784458	High Cost	33.86%
791780.6598	1007634.94	41.688972	92.9481638	Moderate Cost	44.37%
727406.0756	946652.94	39.22804	82.77747083	Moderate Cost	47.39%
727406.0756	861652.1	35.854156	79.5324489	Low Cost	45.08%
715257.2859	781567.8	32.714115	76.59041363	Low Cost	42.71%
714213.9669	781097.56	32.44196	75.92763537	Low Cost	42.73%
714212.9669	847316.75	35.311054	78.90032287	Low Cost	44.75%
714212.9669	936644.6	40.447833	85.56071617	Moderate Cost	47.27%
962286.4045	1208916	53.296064	99.32904692	Moderate Cost	53.66%
791780.6598	962277.44	40.502632	94.2069222	Moderate Cost	42.99%
791780.6598	901015.06	37.320225	82.27810342	Moderate Cost	45.36%
791780.6598	905787.94	36.71128	82.09788474	Moderate Cost	44.72%
585289.3389	790649	32.420498	76.64151363	Low Cost	42.30%
646338.6362	759608.7	31.284641	74.48826244	Low Cost	42.00%
646338.6362	696610.75	28.72143	71.92266047	Low Cost	39.93%

- Transmission Impact represents how much of the Total Energy Cost (LCOE All-in) is contributed by Transmission Cost (LCOT).
- Transmission cost directly affects total energy cost. Even if solar resource is strong, high grid reinforcement can reduce economic feasibility. Therefore, transmission analysis is essential for investment planning.

DAX Measures:

- 681.72K → Sum of Installed Capacity (AC-MW)
- 94.58 → Average Total Energy Cost
- 49.32 → Average Generation Cost
- 3.51 → Average Solar Resource
- 52.65 → Minimum Total Cost



- This dashboard page presents a high-level summary of solar energy potential and cost feasibility under the 2035 Moderate Supply Curve scenario.
- The total installed AC capacity across all sites is **681.72K MW**, indicating strong large-scale solar development potential.
- The **average total energy cost is 94.58 USD/MWh**, which represents the combined generation and transmission cost across all locations.
- The **average generation cost is 49.32 USD/MWh**, showing that production cost alone is moderate. The difference between generation cost and total cost highlights the impact of transmission expenses.
- The **average solar resource is 3.51 kWh/m²/day**, indicating moderate solar irradiance levels suitable for solar power generation.
- The **minimum total cost is 52.65 USD/MWh**, which represents the most economically viable site within the dataset.
- Overall, the summary indicates that while solar resource levels are favorable, transmission infrastructure significantly influences total energy cost.

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- An interactive and consolidated Power BI dashboard was developed to analyse solar resource potential, installed capacity distribution, transmission cost impact, and overall energy cost under the 2035 Moderate Supply Curve scenario.
- The dashboard includes multiple visualizations aligned with the project problem statement and objectives. My Report interactive Layout using Drill-down, Filters and slicers

KPI Cards

- Total Installed Capacity (AC-MW)
- Average Solar Resource
- Average Total Energy Cost
- Minimum Total Energy Cost (Lowest Cost Site)
- Average Generation Cost

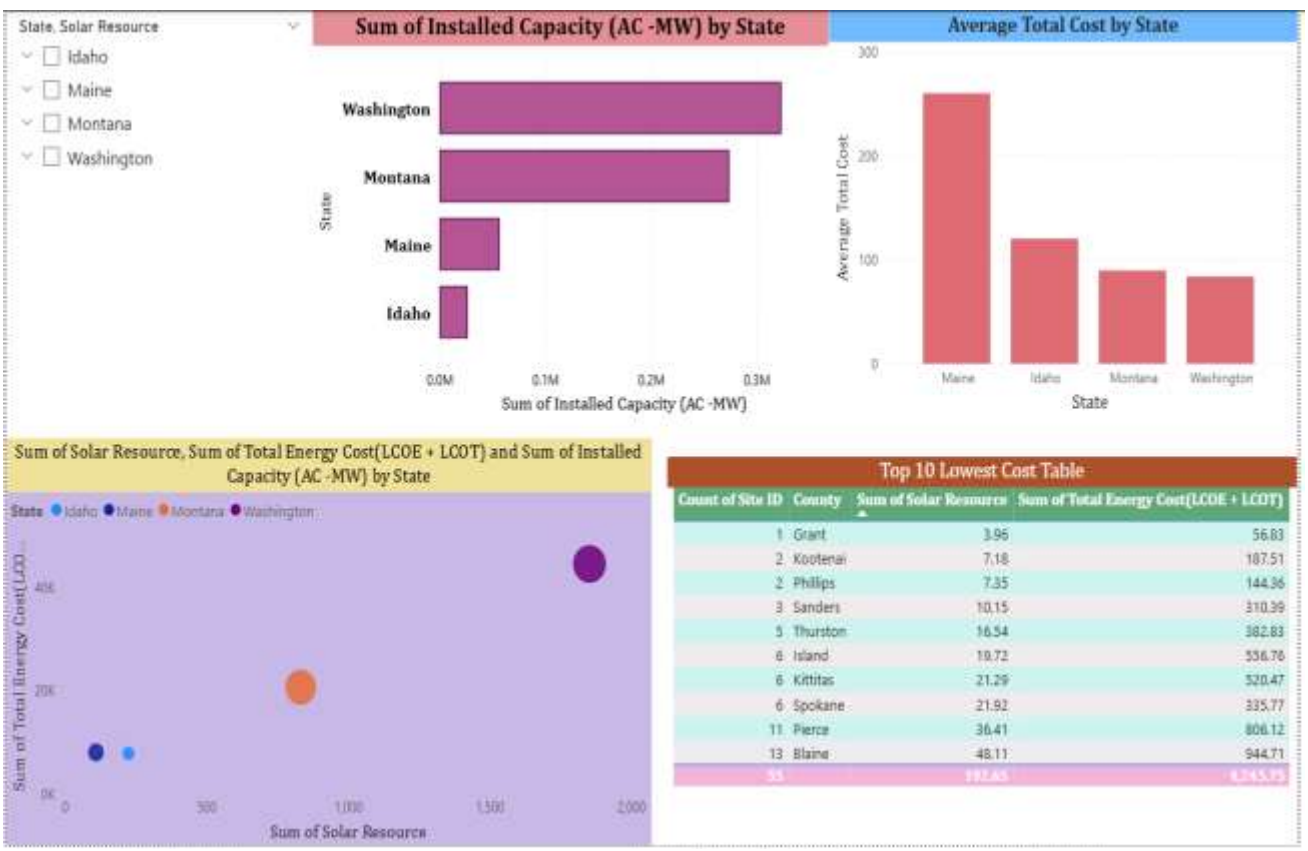
Bar Chart – Installed Capacity by State

Column Chart – Average Total Energy Cost by State

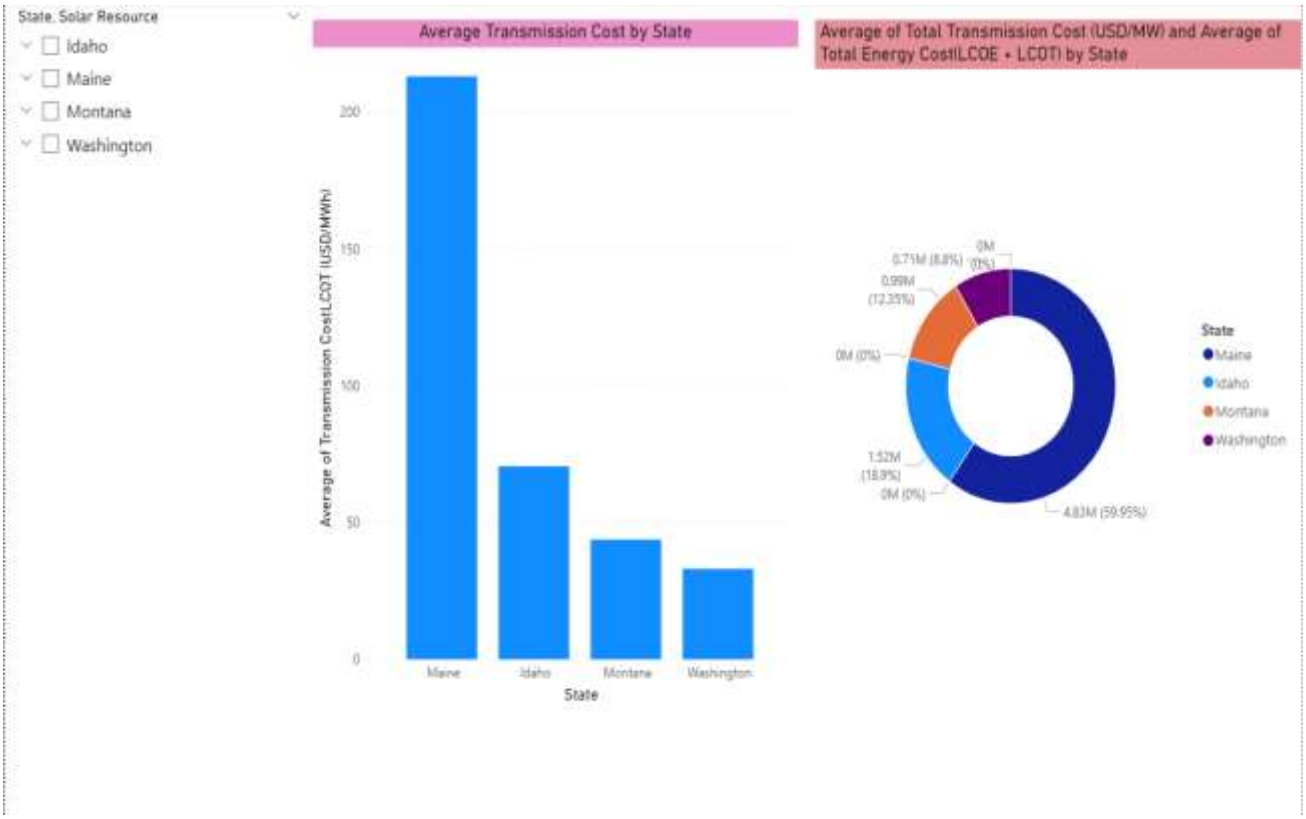
Scatter Chart – Solar Resource vs Total Energy Cost,

Table – Top 10 Lowest Cost Sites

Slicer – Interactive Layout



Count of Site ID	County	Sum of Solar Resource	Sum of Total Energy Cost(LCOE + LCOT)
1	Grant	3.96	56.83
2	Kootenai	7.18	187.51
2	Phillips	7.35	144.36
3	Sanders	10.15	310.39
5	Thurston	16.54	382.83
6	Island	19.72	556.76
6	Kittitas	21.29	520.47
6	Spokane	21.92	335.77
11	Pierce	36.41	806.12
13	Blaine	48.11	944.71
55		192.65	4,245.95



---

9. Insights & Conclusions

- **Key Findings:** Summarize trends, patterns, or anomalies identified in the data

Analysis insights:

Descriptive Analysis:-

- The average solar resource across the selected states is approximately 3.51 kWh/m²/day, indicating moderate solar irradiance levels.
- Washington and Montana show significantly higher installed capacity (AC & DC) compared to Idaho and Maine.
- The overall average total energy cost is 94.58 USD/MWh, with noticeable variation between states.
- Maine has the highest average total energy cost (260.45 USD/MWh), while Washington and Montana have comparatively lower costs.

Diagnostic Analysis:-

- States with higher transmission cost (example: Maine – $213.04) show significantly higher total energy cost.
- Longer grid reinforcement distance increases infrastructure cost, directly impacting LCOT (Levelized Cost of Transmission).
- Higher installed capacity in Washington and Montana indicates better land availability and grid accessibility.
- Regions with lower transmission distance demonstrate better cost efficiency.

Conclusion:-

- Transmission infrastructure plays a major role in increasing total project cost.

Predictive Analysis:-

- If grid reinforcement cost increases further, high-cost states may become less attractive for solar investment.
- States with moderate solar resource but low transmission cost may continue to attract future solar development.
- By 2035, moderate supply curve scenario suggests cost stabilization in regions with strong grid connectivity.

Prescriptive Analysis:-

- **Investment should focus on states with:**
    - ❖ Lower transmission cost
    - ❖ Higher installed capacity
    - ❖ Stable solar resource
- **Priority regions:**
  Washington and Montana (based on lower average total cost and higher capacity)
- Improve grid infrastructure in high-cost states to reduce transmission burden.
- Strategic planning of solar farms closer to grid points can reduce LCOT significantly.

---

10. Conclusions:-

- The analysis indicates that while solar resource availability is important, transmission and grid-related costs significantly influence total energy cost. Regions with strong infrastructure and moderate solar irradiance demonstrate better cost efficiency and investment potential. Therefore, infrastructure optimization is key to sustainable solar expansion under the 2035 moderate supply scenario.